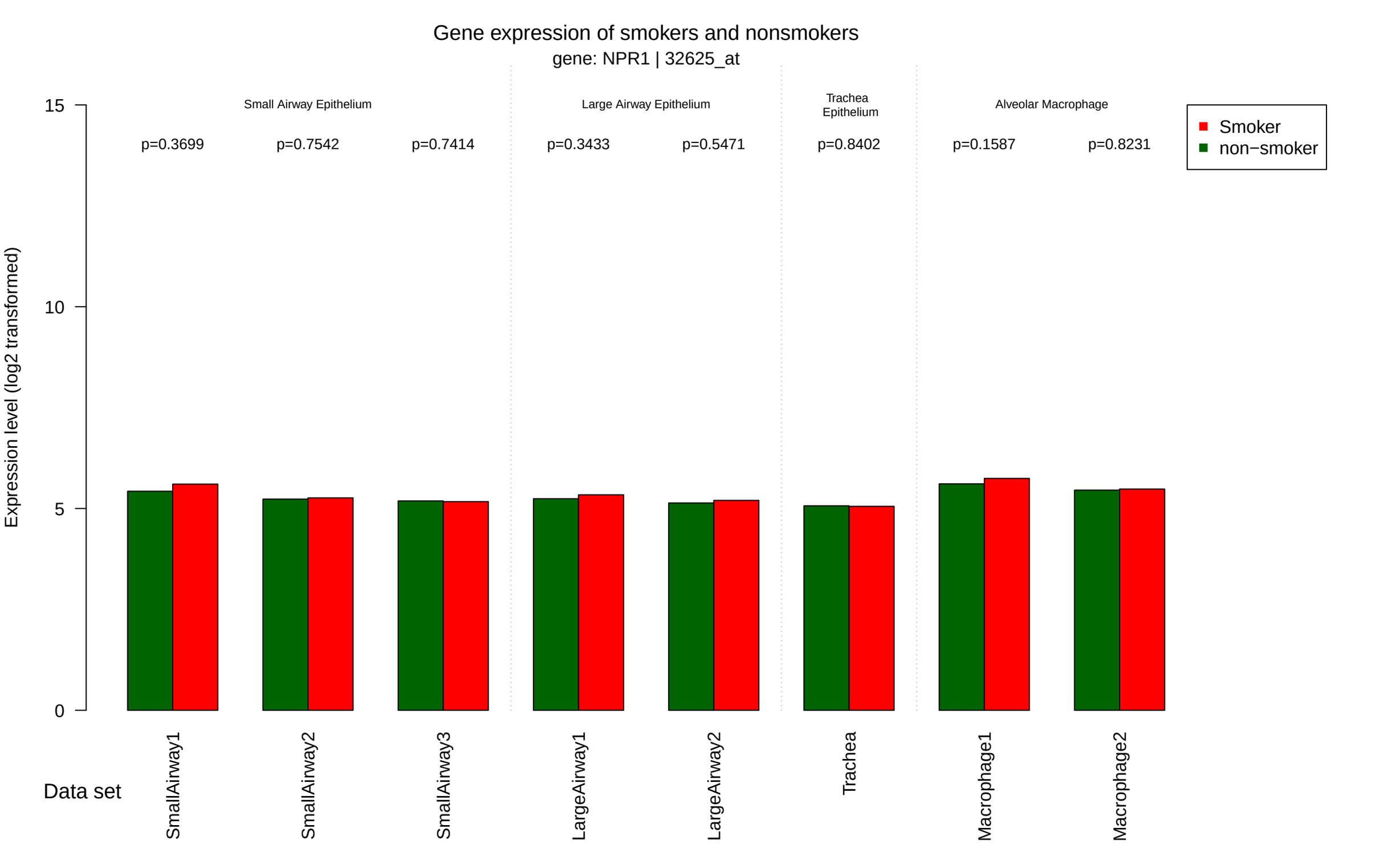
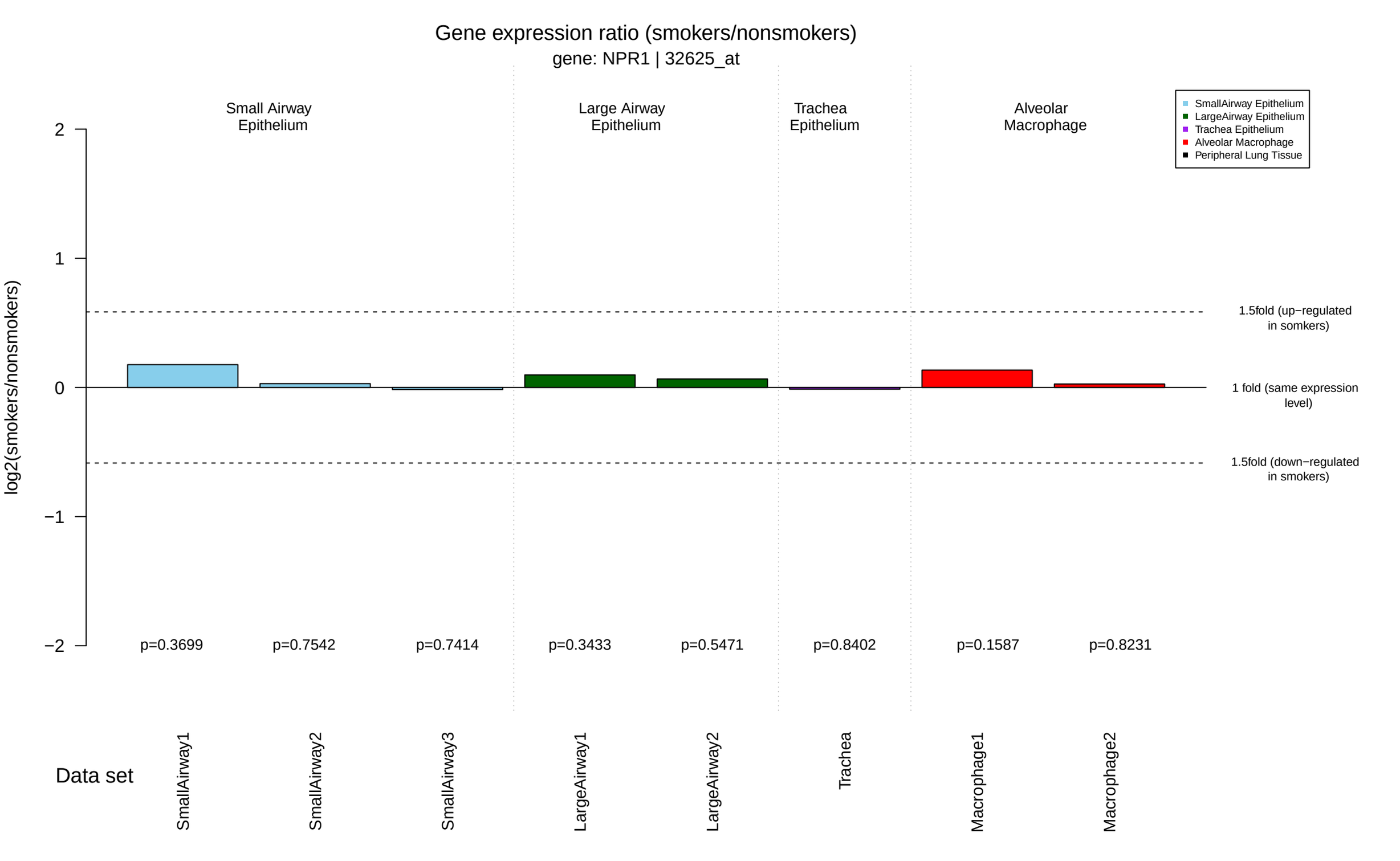
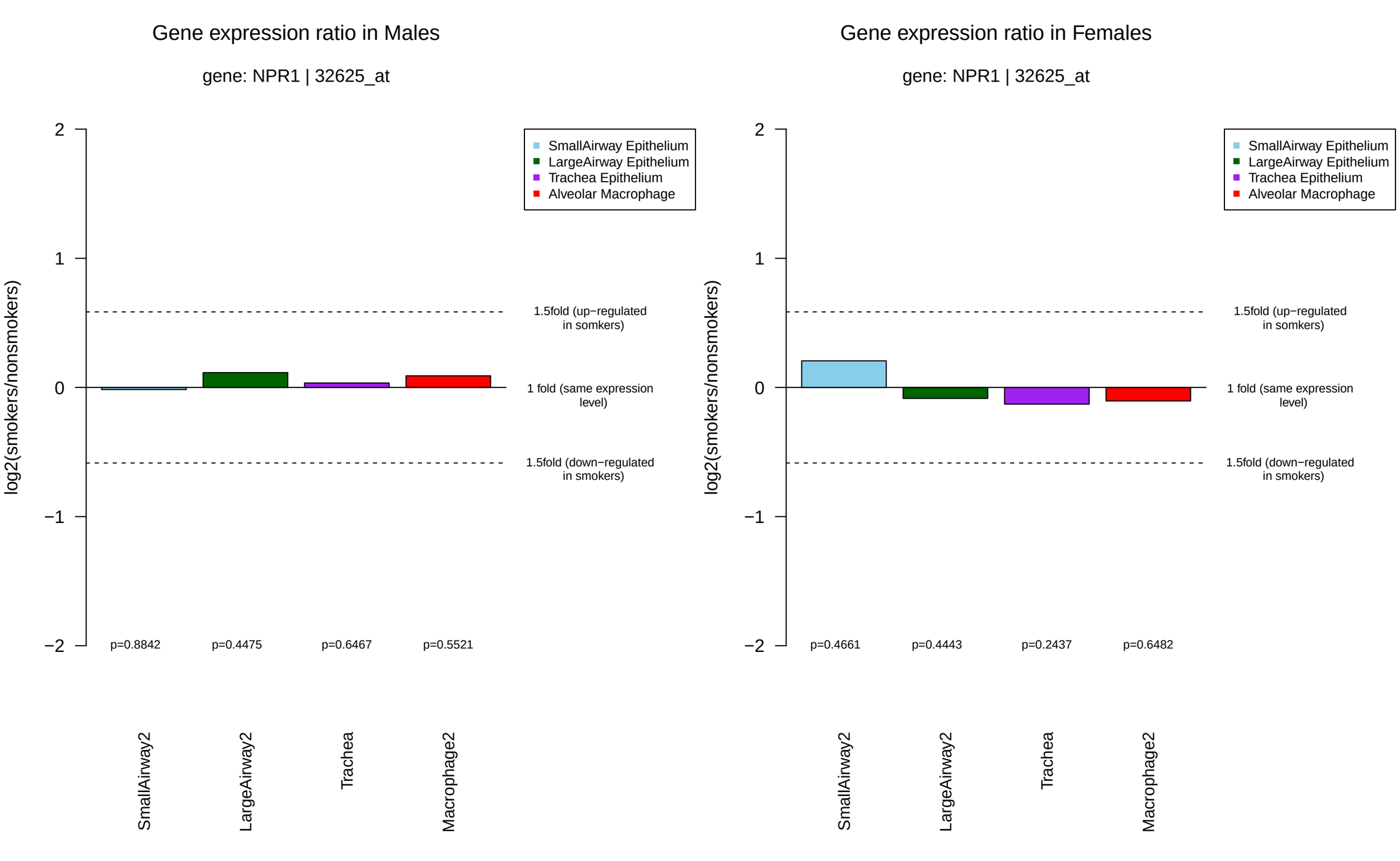
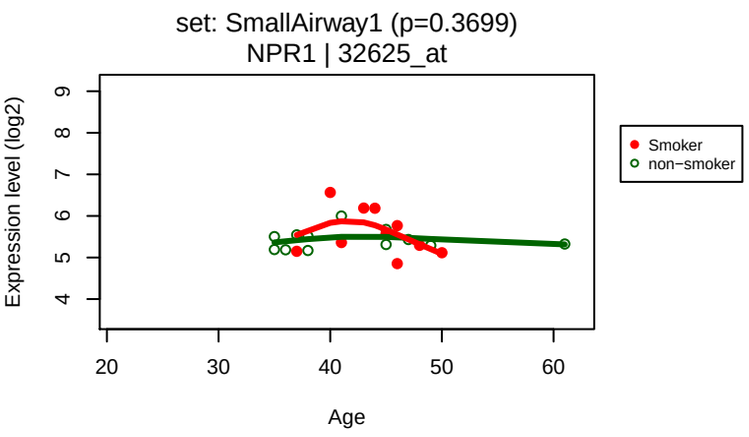
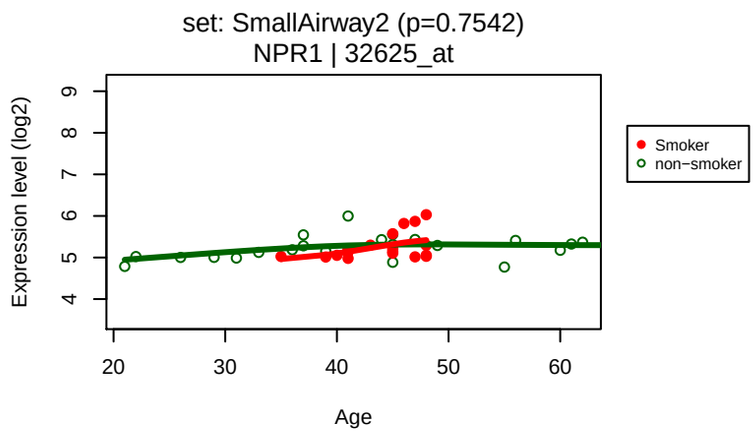
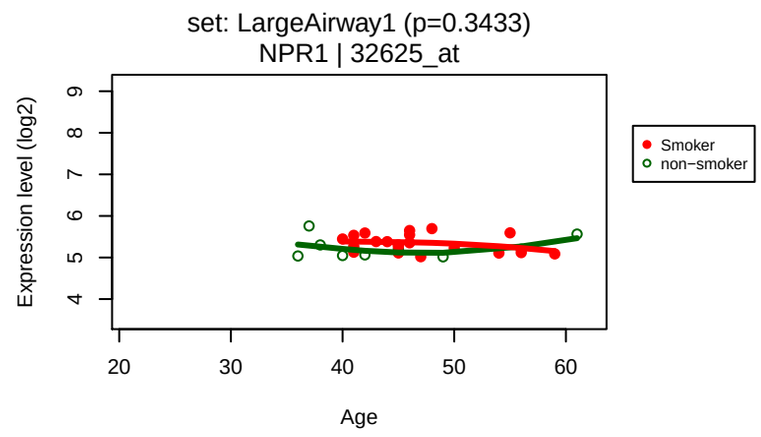

Gene expression of smokers and nonsmokers
gene: NPR1 | 32625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3699
p=0.7542
p=0.7414
p=0.3433
p=0.5471
p=0.8402
p=0.1587
p=0.8231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

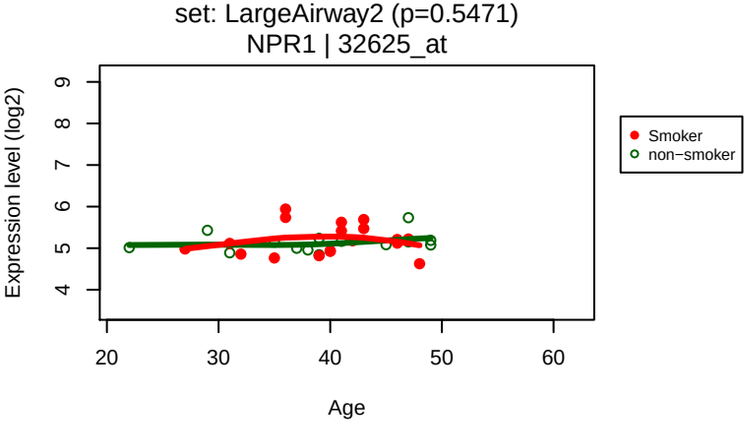

Gene expression ratio (smokers/nonsmokers)
gene: NPR1 | 32625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3699
p=0.7542
p=0.7414
p=0.3433
p=0.5471
p=0.8402
p=0.1587
p=0.8231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

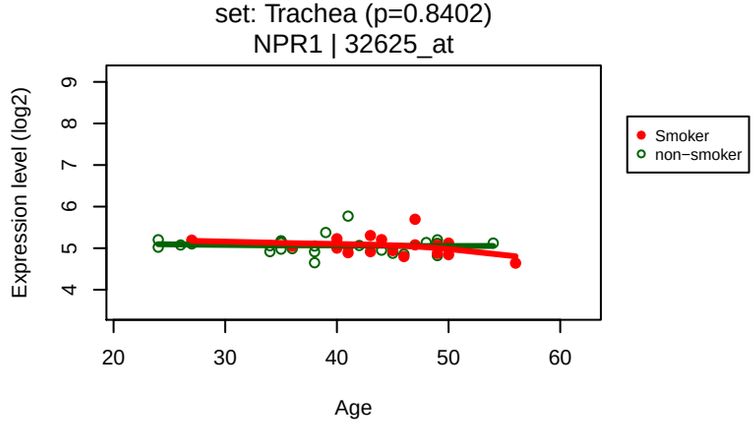

Gene expression ratio in Males
gene: NPR1 | 32625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8842
p=0.4475
p=0.6467
p=0.5521
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPR1 | 32625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4661
p=0.4443
p=0.2437
p=0.6482
SmallAirway2
LargeAirway2
Trachea
Macrophage2

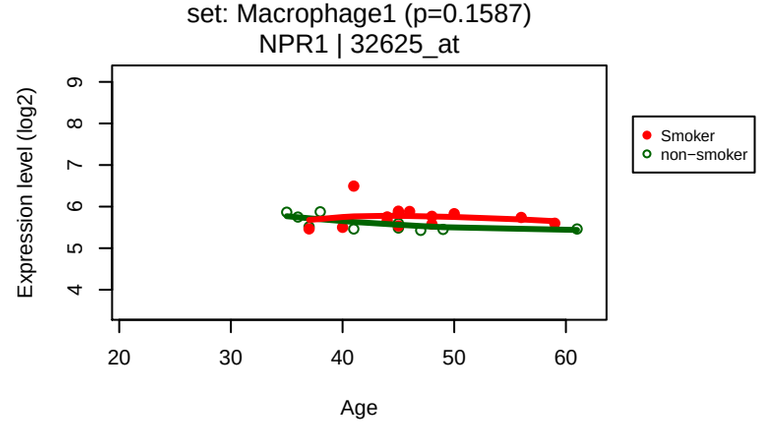

set: SmallAirway1 (p=0.3699)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker

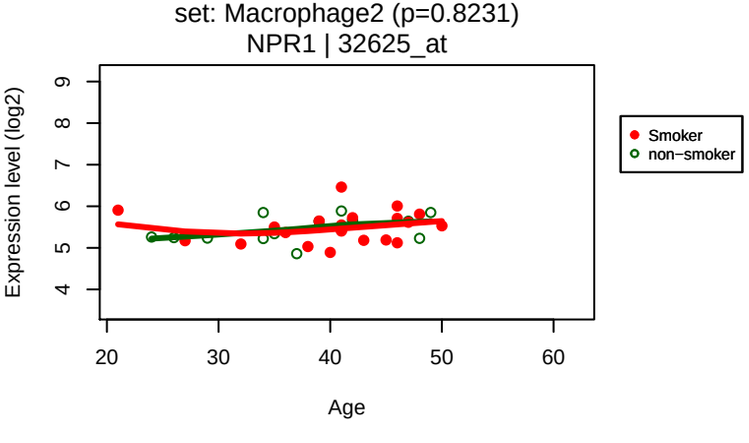

set: SmallAirway2 (p=0.7542)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker

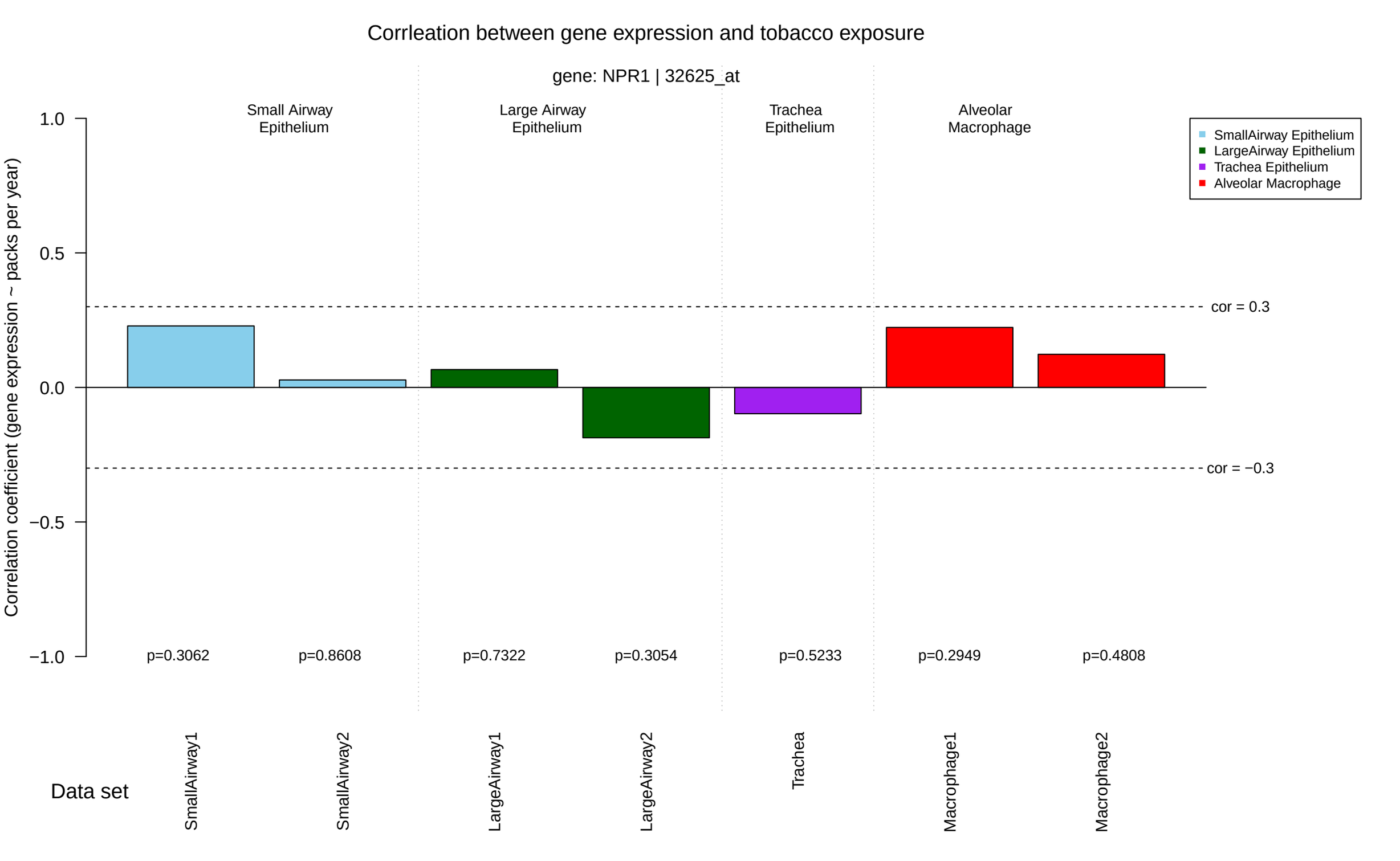

set: LargeAirway1 (p=0.3433)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5471)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8402)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1587)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8231)
NPR1 | 32625_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPR1 | 32625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3062
p=0.8608
p=0.7322
p=0.3054
p=0.5233
p=0.2949
p=0.4808
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2